

Information Geometry

0) Preliminaries

Let M be a manifold. Let $D: M \times M \rightarrow \mathbb{R}_{\geq 0}$. For $p \in M$, let

$$g_p = \partial_p^* D: T_p M \oplus T_p M \rightarrow \mathbb{R}.$$

Then D is a **divergence** if

$$D(p, q) \geq 0 \quad \forall p, q \in M \quad \text{with } D(p, q) = 0 \text{ iff } p = q,$$

and g_p is positive-definite. note that it is automatically symmetric

Note: We have the following symmetries that follow from $D(p, q) \geq 0$ with equality only on the diagonal.

First order derivatives vanish because $D(p, \cdot)$ and $D(\cdot, p)$ have (global)

$$(I) \text{ minima at } p: \quad i^*(\partial_1 D) = i^*(\partial_2 D) = 0 = i^*(\partial D) \quad \leftarrow \text{diagonal embedding } i: M \hookrightarrow M \times M$$

For $X \in TM$, let $X_1, X_2 \in TM \times TM$. Then

$$(II) \quad i^*(X_1, Y_1 D) = i^*(X_2, Y_2 D) = -i^*(X_1, Y_2 D)$$

by differentiating (I). Note that this defines the associated Riemannian metric.

We define the **primal connection** ∇ by

$$g(\nabla_X Y, Z) = -i^*(X_1, Y_1 \partial_2 Z D),$$

and the **dual connection** ∇^* by

$$g(\nabla_X^* Y, Z) = -i^*(X_2, Y_2 \partial_1 Z D).$$

Note: Both the primal and the dual connections are **torsion-free**, i.e.

$$T(X, Y) := \nabla_X^* Y - \nabla_Y^* X - [X, Y] = 0.$$

Note: These connections are dual in the sense of

$$Z g(X, Y) = g(\nabla_Z X, Y) + g(X, \nabla_Z^* Y).$$

In particular, the Levi-Civita connection ∇^{LC} is decomposed as

$$\nabla^{LC} = \frac{1}{2}(\nabla + \nabla^*).$$

Example: Let M be a Riemannian manifold. Then locally, the distance function d is a divergence. It induces the Riemannian metric and its primal and dual connections equal the Levi-Civita connection.

Example: Let $\Psi: \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function. Then

$$D_\Psi[z: z_0] := \Psi(z) - \Psi(z_0) - \nabla \Psi(z_0) \cdot (z - z_0),$$

the distance of the graph of Ψ at z to the hyperplane tangent to the graph at z_0 , is a divergence, called the **Bregman divergence**.

Example (Euclidean): For $\Psi(z) = \frac{1}{2} \sum z_i^2$, the Bregman divergence is the Euclidean distance.

Example (Kullback-Leibler): Let $\Psi(z) = \sum z_i \log z_i$ be the (negative) Shannon entropy. Then the Bregman divergence is the **Kullback-Leibler divergence**

$$D[z: z'] = \sum (z_i \log \frac{z_i}{z'_i} - z_i + z'_i) \quad \leftarrow \text{zero for probability measures}$$

Example (Free energy of an exponential family): Let $p: \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$ be

$$p(x, \theta) = \exp(\sum \theta_i x_i + k(x) - \Psi(\theta)),$$

such that $\int p(x, \theta) dx = 1$. We view the map $\theta \mapsto p(\cdot, \theta)$ as a coordinate chart on the set of probability distributions. Note that the **free energy**

$$\Psi(\theta) = \log \int \exp(\sum \theta_i x_i + k(x)) dx$$

is convex. Differentiating $\int p(x, \theta) dx = 1$ w.r.t. θ , we get

$$\nabla \Psi(\theta) = \mathbb{E}[x].$$

Hence, Ψ induces the KL divergence

$$D[p(\cdot, \theta): p(\cdot, \theta')] = \int p(x, \theta') \log \frac{p(x, \theta')}{p(x, \theta)} dx.$$

Let $\Psi: \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function. Consider Ψ as a function on M . Let $\partial \Psi: M \rightarrow T^*M$ have image $\Omega = \text{Im}(\partial \Psi) \subset T^*M$. Then the map

$$\mathbb{L}_\Psi = \partial \Psi: M \xrightarrow{\sim} \Omega$$

is the **Legendre transformation** of Ψ . In local coordinates,

$$(z_1, \dots, z_n) \mapsto \left(\frac{\partial \Psi}{\partial z_1}, \dots, \frac{\partial \Psi}{\partial z_n} \right).$$

The Legendre transform is a bijection between the graph of Ψ and its tangent hyperplanes.

We a function $\Psi^*: \Omega \rightarrow \mathbb{R}$ by

$$\Psi^*(z^*) = g(z, z^*) - \Psi(z),$$

where $z = \mathbb{L}_\Psi^{-1}(z^*)$.

The function Ψ can be recovered from the set of tangent hyperplanes together with their z -axis intercepts Ψ^*

Note ($\Psi^{} = \Psi$):** We have

$$\partial \Psi^*(v) = g(\partial z(v), z^*) + g(z, \partial z^*(v)) - \partial \Psi(v) = g(z, v),$$

i.e. $\partial_z \Psi^* = z = \mathbb{L}_\Psi^{-1}(z^*)$.

Note: The Bregman divergence of Ψ^* equals that of Ψ .

$$D_{\Psi^*}[z^*: z'^*] = D_\Psi[z: z'].$$

Note: We have

$$D_\Psi[P: Q] = \Psi(z_P) + \Psi^*(z_Q^*) - z_P \cdot z_Q^*$$

Let ∇, ∇^* be dual connections on a Riemannian manifold M , i.e.

$$X g(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X^* Z).$$

We call (M, ∇, ∇^*) **dually flat** if the Riemann curvature of ∇ (and equivalently of ∇^*) vanishes:

$$R(X, Y)Z := \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z = 0$$

Note: Generally, we have

$$g(R(X, Y)Z, W) = -g(Z, R^*(X, Y)W).$$

Proposition (Dual flatness): The following are equivalent:

I) M is dually flat;

II)